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AutoMedic: An Automated Evaluation Framework for Clinical Conversational Agents with Medical Dataset Grounding

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***Abstract*— Evaluating large language models (LLMs) has recently emerged as a critical issue for safe and trustworthy application of LLMs in the medical domain. Although a variety of static medical question-answering (QA) benchmarks have been proposed, many aspects remain underexplored, such as the effectiveness of LLMs in generating responses in dynamic, interactive clinical multi-turn conversation situations and the identification of multi-faceted evaluation strategies beyond simple accuracy. However, formally evaluating a dynamic, interactive clinical situation is hindered by its vast combinatorial space of possible patient states and interaction trajectories, making it difficult to standardize and quantitatively measure such scenarios. Here, we introduce *AutoMedic*, a multi-agent simulation framework that enables automated evaluation of LLMs as clinical conversational agents. AutoMedic transforms off-the-shelf static QA datasets into virtual patient profiles, enabling realistic and clinically grounded multi-turn clinical dialogues between LLM agents. The performance of various clinical conversational agents is then assessed based on our *CARE* metric, which provides a multi-faceted evaluation standard of clinical conversational accuracy, efficiency/strategy, empathy, and robustness. Our findings, validated by human experts, demonstrate the validity of AutoMedic as an automated evaluation framework for clinical conversational agents, offering practical guidelines for the effective development of LLMs in conversational medical applications.**

***Index Terms*— Large Language Models, Agentic AI, Automated Evaluation, Conversational Agents, Clinical Simulation, Multi-agent Systems**

I. Introduction

THE rapid advancement of large language models (LLMs) has spurred the development of sophisticated foundation models across both proprietary and open-source landscapes [1]–[8]. This has also led to a significant interest in their medical applications, resulting in a variety of specialized models fine-tuned on clinical data [9]–[15] and the emergence of chatbots designed for direct human interaction [16]–[19]. Notably, experimental evidence suggests that the medical knowledge of these LLMs on standard medical question-answering (QA) benchmarks has already reached a level comparable to that of human experts [20].

Despite these impressive benchmark scores, static evaluations based on existing medical QA benchmarks primarily focus on information retrieval rather than the dynamic and interactive nature of real-world clinical encounters. While current models possess the technical capacity for human interaction, they cannot be safely or effectively integrated into clinical workflows without an evaluation framework that rigorously validates their performance in interactive scenarios. Two significant areas for further development exist in this context. First, the evaluation process in standard benchmarks is inherently static. Unlike a real clinical care scenario, where the clinician aims to fulfill a professional role by interacting with the patient to elicit information, these benchmarks provide no measure of conversational and diagnostic inquiry skills. Effective patient-doctor communication is fundamental to clinical care, as it serves as the primary method for gathering nuanced patient history, building trust, and ensuring patient adherence to treatment plans. An agent's ability to communicate with empathy and clarity is not just a feature but a core requirement for safe and effective deployment. Furthermore, given the susceptibility of LLMs to user-driven misinformation

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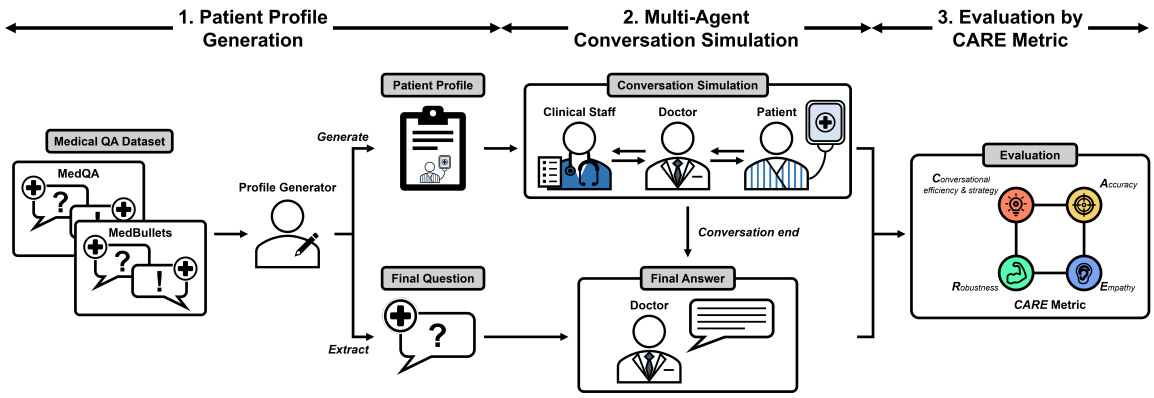


Fig. 1. Overview of the proposed AutoMedic framework. AutoMedic framework automatically converts a medical QA dataset into a virtual patient scenario, facilitates a multi-agent clinical simulation, and evaluates the LLM agent's performance based on our proposed metrics.

or leading inputs [21], [22], the ability to maintain robust and accurate dialogue is critical for safe clinical deployment. Second, the queries in these benchmarks are typically self-contained, providing all the necessary information upfront. In actual clinical practice, however, critical information must be actively gathered through patient dialogue and diagnostic tests, which represents a complex interactive task.

Evaluating LLMs as clinical conversational agents is an established and essential practice within the medical AI community to ensure their effectiveness in dynamic environments. To address the inherent constraints of traditional static benchmarks, several recent studies have developed frameworks specifically designed to assess these models within interactive clinical scenarios [23]–[28]. These frameworks have demonstrated the feasibility of evaluating LLMs as clinical conversational agents by introducing novel datasets and simulating clinical scenarios. However, achieving full automation remains a key objective because existing workflows 1) require specially designed datasets, 2) necessitate substantial human labor or rely on non-realistic clinical simulations, and 3) heavily rely on human evaluators for key qualitative metrics. Moreover, these methods frequently focus on diagnostic scenarios and would benefit from clearer quantitative metrics.

To address these shortcomings, this paper introduces *AutoMedic*, a fully automated framework for evaluating LLMs as clinical conversational agents in clinical care scenarios, with the overall scheme illustrated in Fig. 1. Our key contributions are:

- **An automatic method for converting static medical QA datasets into interactive virtual patient profiles.** This approach provides a scalable methodology for transforming standard off-the-shelf QA benchmarks into conversational evaluation suites, which reduces the dependency on specialized and hand-crafted datasets. Our framework incorporates a filtering mechanism to identify and select the subset of source QA items most suitable for realistic dialogue.
- **A multi-agent simulation framework that models a realistic clinical dialogue.** In this framework, the LLM-based clinical conversational agent must actively gather information by conversing with a virtual patient and requesting test results from a clinical staff agent before making a final judgment. The entire evaluation process is fully automated, requiring no human intervention.
- **A multi-faceted metric for quantitative evaluation.** We propose the *CARE* metric to assess performance across four key rubrics: accuracy, conversational efficiency and strategy, empathy, and robustness. By integrating and quantifying these crucial aspects within a single automated framework, our metric offers a comprehensive guideline for selecting and developing LLMs for assistive clinical roles.

The remainder of this paper is organized as follows. Section II reviews related works on medical LLMs and conversational agent evaluation. Sections III, IV detail our proposed framework and experimental setup. Section V and Section VI present the experimental results and discussion, and Section VII concludes the paper.

II. Related Works

A. LLMs for the Medical Domain

Leading LLMs have consistently demonstrated high performance on a range of medical tasks and benchmarks. OpenAI's GPT-4, for example, surpassed the passing score of the United States Medical Licensing Examination (USMLE) by over 20 points [29]. Its successor, GPT-5 [8], continued this trend by achieving the highest score to date on the HealthBench benchmark [25]. Similarly, DeepSeek-R1 [5] has shown strong performance in clinical decision-making tasks [30] and has achieved the top win-rate on several medical benchmarks, outperforming other prominent proprietary and open-source LLMs [31].

Beyond the aforementioned general-purpose LLMs, a significant amount of research has focused on developing LLMs specifically tailored for the medical domain. Models such as HuatuoGPT-o1 [10] and m1 [32] are fine-tuned on medical problem sets to enhance their clinical reasoning abilities. In parallel, Google has introduced a series of powerful multimodal medical LLMs, including Med-PaLM M [33], Med-Gemini [34], and MedGemma [15]. These models have demonstrated exceptional performance across a diverse range of medical benchmarks.

However, since most current evaluations primarily assess the knowledge of LLMs via simple question-answering tasks,

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concerns arise regarding their ecological validity. Specifically, these methods may fail to adequately gauge a model's utility in real-world clinical settings, which inherently necessitate explorative information gathering for accurate diagnosis and treatment planning [35]–[39]. To address these limitations, recent research [19] has focused on training LLMs as specialized diagnostic agents capable of managing complex, multi-turn diagnostic processes.

B. Evaluation of Clinical Conversational Agents

Several pioneering efforts have sought to evaluate LLMs as clinical conversational agents by creating specialized benchmarks and systems. For instance, Google's Articulate Medical Intelligence Explorer (AMIE), a system for diagnostic dialogue, was evaluated through interactions with human actors playing the role of simulated patients [28]. Similarly, OpenAI developed HealthBench, a benchmark comprising 5,000 realistic health conversations created in collaboration with physicians, complete with detailed scoring rubrics [25]. More recently, Microsoft introduced the MAI Diagnostic Orchestrator (MAI-DxO), a system where multiple LLM agents collaboratively debate to make an accurate diagnosis. To assess it, they also released the Sequential Diagnosis Benchmark (SDBench), derived from complex clinicopathological cases from the NEJM [27]. Furthermore, Qiu et al. [40] introduced MedR-Bench, a collection of 1,453 patient cases accompanied by reference reasoning, to evaluate the diagnostic reasoning of LLMs using a specialized Reasoning Evaluator. A common limitation of these approaches is their dependence on bespoke datasets. The creation of such resources is a time-consuming and labor-intensive process, which helps explain why the number of conversational benchmarks remains limited compared to the abundance of static medical QA datasets. Moreover, these datasets are often limited in conversational depth, lacking a significant number of multi-turn exchanges. Instead of serving as active agents that drive the dialogue, models in these settings are relegated to passive roles, evaluating fixed conversation histories rather than engaging in actual multi-turn interactions.

To overcome these limitations, several studies have specifically focused on evaluating the interactive capabilities of LLM agents in patient dialogue. A notable example is AgentClinic, a multimodal agent benchmark for evaluating LLMs in simulated clinical environments [23]. The AgentClinic framework generates virtual patient and measurement agents from medical QA data. A doctor agent must then interact with these simulated agents to gather the necessary information for a medical decision. The authors use this simulation to compare the diagnostic accuracy of LLM agents across various biases and languages, supplementing the results with human evaluations of the conversation quality. Similarly, MedAgentSim [24] simulates doctor-patient interactions but distinguishes itself by utilizing multiple doctor agents to collaboratively derive the final clinical decision. Furthermore, the CRAFT-MD framework [41] evaluates the performance of clinical LLMs by situating them within a simulated interactive environment to assess their diagnostic accuracy. Despite their comprehensive approach, these approaches have two key limitations relevant to our work. First, it does not provide a method for assessing whether leveraging an existing medical QA benchmark is suitable for conversion into a realistic conversational simulation. Second, it lacks a quantitative framework for automatically evaluating an agent's performance across multiple conversational aspects, relying instead on subjective manual human assessment or solely on the accuracy of the QA task.

III. AutoMedic Framework

The AutoMedic framework, illustrated in Fig. 1, is composed of three primary stages: 1) patient profile generation, 2) multi-agent conversation simulation, and 3) automated evaluation. This entire process is orchestrated by four specialized LLM agents: a profile generator, a doctor, a patient, and a clinical staff. The following sections provide the specific roles of these agents and a detailed description of each stage.

A. Agents

The AutoMedic framework operates using four distinct LLM agents, each with a specialized role. The primary agent is the *doctor* agent, which emulates a physician and serves as the sole subject of our evaluation. The other three agents, the *profile generator*, *patient*, and *clinical staff*, act in supporting roles to create the environment for this evaluation.

The process begins with the profile generator, which performs two sequential tasks. First, it filters a set of medical QA queries to select items suitable for simulation. Second, it transforms each valid query into a structured virtual patient profile.

This profile is then used by the patient agent, which instantiates the patient's persona. It dynamically responds to inquiries about symptoms and medical history, and can adapt to a caregiver role in scenarios where a patient cannot self-represent (e.g., due to age or cognitive impairment).

To simulate the broader clinical environment, the clinical staff agent acts as a gatekeeper for technical data. When the doctor agent requests a test, this agent provides the corresponding results from the profile. To ensure factual grounding, it only returns pre-existing information and will state that a result is unavailable if it is not in the profile, preventing data hallucination.

Within this environment, the doctor agent's objective is to solve the medical problem by strategically gathering information from both the patient and clinical staff agents. For baseline comparison, this agent also directly answers the original static QA query without any interaction.

B. Patient Profile Generation

The first stage of the AutoMedic framework is patient profile generation (Fig. 2(a)). This process transforms a static medical QA query into a structured virtual patient profile, which then serves as the foundation for the subsequent conversation simulation. The process begins when the profile generator agent receives a medical QA query, comprising a medical context, a question, several options, and a correct answer, and

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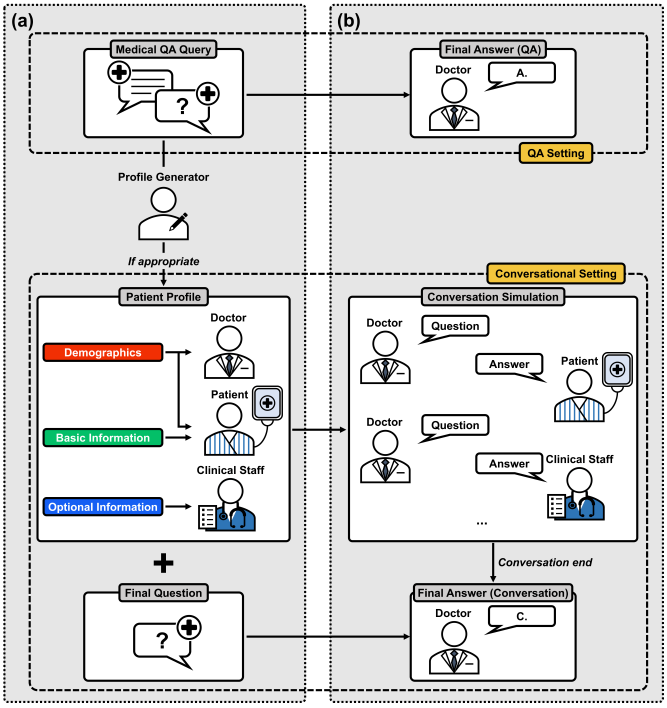


Fig. 2. The AutoMedic framework, detailing (a) the automatic generation of a virtual patient profile from a medical QA query, and (b) the subsequent multi-agent conversation simulation.

first assesses its suitability. As our framework is designed for patient-specific decision-making scenarios, the following types of queries are automatically filtered out and excluded from the dataset by the profile generator agent:

- Queries describing research participants or experimental settings rather than clinical care.
- Those focusing on abstract concepts (e.g., pathophysiology, molecular mechanisms) instead of a specific patient case.
- General knowledge or fact-recall questions (e.g., definitions, classifications).
- Image-dependent queries that do not describe the visual findings in the text.
- Queries where critical patient details (e.g., age, chief complaint) are only present in the answer choices.

Once a query is determined to be suitable, the profile generator agent proceeds to create the virtual patient profile. It extracts relevant information from the medical context and organizes it into three distinct categories:

- **Demographics**: Basic patient data such as age and sex.
- **Basic Information**: Core clinical details including the chief complaint, duration of symptoms, family and past medical history, and relevant lifestyle factors (e.g., smoking, alcohol use).
- **Optional Information**: Data that may not be present in all cases, such as vital signs, laboratory tests, physical examinations, imaging results, and pathology results.

To enhance the realism of the simulation, the profile generator is instructed to impute any missing basic information items with plausible, randomly generated values, ensuring these additions do not alter the correct answer to the original query. Conversely, optional information is never generated if it is absent. This is a crucial constraint, as many queries test the doctor agent's ability to determine which tests are necessary. Generating these results upfront would interfere with the evaluation. Finally, the agent isolates the specific medical question and its corresponding response options from the original query, excluding the clinical vignette. This extracted content is presented to the doctor agent upon the conclusion of the conversation simulation.

Once the patient profile is generated, the information is selectively distributed among the agents according to their role, to simulate realistic knowledge boundaries (Fig.2(b)). The doctor agent receives only the demographics as the baseline information of the given patient. The patient agent is provided with both demographics and basic information, enabling it to accurately represent the patient's history and symptoms to answer the query from the doctor agent. To maintain a realistic simulation, the patient agent provides specific details only when directly prompted by the doctor agent. It is explicitly restricted from disclosing objective medical data such as laboratory results or imaging reports, as these are typically unavailable to a patient in a clinical setting. Furthermore, if the doctor agent requests information outside this defined scope, the patient agent is programmed to respond naturally by stating that it does not have that specific data, which closely mirrors real-world patient behavior.

Lastly, the clinical staff agent serves a distinct role by managing objective medical data such as laboratory results and physical examination findings. Unlike the patient agent, who provides subjective information based on symptoms and personal history, the clinical staff agent only discloses technical data when explicitly prompted by the doctor agent. By isolating this technical information until specifically requested, this design maintains the simulation's integrity and ensures the proper evaluation of the doctor agent's information-gathering skills. This separation is critical for the evaluation process because it forces the doctor agent to demonstrate proactive diagnostic reasoning rather than passively receiving a complete case summary. Consequently, the clinical staff agent acts as a gatekeeper for the data necessary to reach a correct diagnosis, which allows us to measure the model's ability to request the appropriate clinical evidence at the correct stage of the consultation.

C. Multi-Agent Conversation Simulation

The multi-agent conversation simulation begins after the patient profile has been generated, as illustrated in Fig. 2(b). The doctor agent initiates and directs the entire dialogue using specific tags. Questions for the patient agent are wrapped in `<patient></patient>` tags, while requests for the clinical staff agent are framed within `<clinical></clinical>` tags. The patient and clinical staff agents automatically respond to any text directed to them within these respective tags.

This interactive process continues until one of two termination conditions is met: either the doctor agent concludes the dialogue by issuing an `</end>` tag, or the conversation reaches a predefined maximum length. We define a single

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turn as a query from the doctor agent and the corresponding response from another agent. For our experiments, we set the maximum number of turns to 20, which is sufficient if the doctor agent gathers relevant information efficiently.

Upon the conversation's conclusion, the doctor agent is presented with the final medical question that was extracted from the original query. The doctor agent then formulates an answer based solely on the information it gathered during the simulated dialogue. To establish a performance baseline, we also collect responses in a standard QA setting. As depicted at the top of Fig. 2, the doctor agent is presented with the complete original medical query, containing full patient context upfront. In this static configuration, the agent must generate an answer without any conversational interaction, mirroring traditional QA tasks. Both the conversational and static QA responses are subsequently compared during the evaluation phase.

D. Automated Evaluation

The final stage of the AutoMedic framework is the automated evaluation of the doctor agent. This evaluation is based on the entire transcript of the multi-agent conversation as well as the final answer provided. To conduct this assessment automatically and without human intervention, we introduce the *CARE* (Conversation efficiency and strategy, Accuracy, Robustness, and Empathy) metric. This metric quantitatively assesses the agent's performance across the following four distinct aspects:

1. Accuracy: The primary goal of this score, S_{ACC}, is to evaluate the diagnostic accuracy of the doctor agent, while also accounting for any performance degradation that occurs when moving from a static to a conversational setting. This formulation is grounded in the clinical requirement for knowledge retention under cognitive load. In real-world practice, a clinician's value is defined not just by their static medical knowledge but by their ability to apply that knowledge consistently during the complexities of a patient interaction. Many LLMs exhibit a performance collapse when moving from isolated benchmarks to multi-agent dialogues, indicating a lack of conversational robustness.

First, we independently calculate the accuracy for the conversational setting (Acc_{Conv}) and the static QA setting (Acc_{QA}):

$$Acc_{\text{Conv}} = \frac{n_{\text{Conv}}}{N}, \quad Acc_{\text{QA}} = \frac{n_{\text{QA}}}{N}, \tag{1}$$

where N is the total number of queries, and n_{Conv} and n_{QA} are the number of correct answers in the conversational and static QA settings, respectively.

The final accuracy score, S_{ACC}, is then defined as:

$$S_{\text{ACC}} = Acc_{\text{Conv}} \times \frac{Acc_{\text{Conv}}}{Acc_{\text{QA}}}. \tag{2}$$

This formula incorporates two components. The first term, Acc_{Conv}, directly measures the agent's performance in the interactive simulation. The second term, $Acc_{\text{Conv}}/Acc_{\text{QA}}$, acts as a retention-based penalty factor. It quantifies the fidelity of the agent's reasoning by measuring how much of its baseline medical knowledge is preserved during the dialogue. An agent that maintains high accuracy in the conversation relative to its baseline QA performance will score higher. This ensures that the metric rewards not only the final correct answer but also the model's ability to maintain clinical reasoning integrity throughout the information-gathering process.

2. Conversational Efficiency & Strategy: This score, S_{CES}, evaluates the doctor agent's conversational efficiency and its adherence to a natural, strategic dialogue flow. Our formulation is grounded in the Smith model of patient-centered interviewing [42], which emphasizes a structured transition from open-ended inquiry to focused, non-repetitive questioning. In real clinical encounters, effective communication requires a balance between gathering sufficient data and minimizing patient burden. Physicians typically ask one or two focused questions at a time, iteratively building a diagnostic hypothesis based on patient responses. Conversely, some LLM agents may adopt an unnatural "checklist" approach, asking numerous questions simultaneously to gather information quickly. While this might appear efficient in terms of raw information acquisition, it violates the clinical principle of focused inquiry and is not representative of a professional diagnostic strategy.

To quantify this behavior, we employ a Gaussian-based metric that rewards adherence to an ideal clinical cadence, reflecting the Calgary-Cambridge guide's [43] standards for flow and structures. We define q_{turn} as the average number of sentences per doctor agent request, and T as the total number of conversation turns. For each sample i, the conversational efficiency and strategy score, s^i_{CES}, is calculated as follows:

$$s^i_{\text{CES}} = \left[\exp\left(-\frac{(q_{\text{turn}} - \mu_q)^2}{2\sigma_q^2}\right) \cdot \exp\left(-\frac{(T - \mu_T)^2}{2\sigma_T^2}\right)\right] \cdot P_{\text{ACC}}. \tag{3}$$

In this formulation, μ_q and σ_q represent the expected average and standard deviation for sentences per turn, while μ_T and σ_T serve the same roles for the total number of turns. P_{ACC} is a soft accuracy penalty that is set to 1 when the final answer is correct, and 0.5 when it is incorrect. The parameters μ_q, σ_q, μ_T, and σ_T are adjustable, allowing the framework to be calibrated against different configurations or clinical contexts. This scoring mechanism penalizes deviations from professional norms. For example, a doctor agent receives a low score for information dumping (asking too many questions in a single turn) or for being overly passive. Similarly, the score decreases if the agent consumes excessive turns inefficiently or concludes the encounter prematurely without sufficient interaction.

The final S_{CES} score is then computed as the scaled average across all N samples:

$$S_{\text{CES}} = \frac{1}{N}\sum_{n=1}^{N} s^i_{\text{CES}}. \tag{4}$$

3. Empathy: The third score, S_{EMP}, provides an approximation of the empathy perceived by the patient agent during the simulation. Since LLMs can sometimes interact in a robotic or emotionally detached manner, evaluating their ability to convey empathy is crucial for patient-centric applications. This

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metric is theoretically grounded in the relationship-centered care model [44], which positions emotional rapport as a technical necessity for clinical success rather than a mere social nicety. Furthermore, our evaluation criteria align with the SEGUE Framework [45], specifically focusing on the doctor's ability to elicit the patient's perspective and acknowledge their emotional state. Recent studies have emphasized the importance of measuring empathetic capabilities of clinical LLMs, particularly their ability to provide supportive and human-centric interactions [28], [46], [47].

To quantify this, we use the patient agent as a proxy evaluator. This approach mirrors real-world healthcare quality assessment where patient satisfaction surveys are used to gauge provider performance. After each conversation concludes, the patient agent is prompted to rate the doctor agent's empathy on a 5-point scale (where 1 is low and 5 is high), a standardized assessment format widely adopted in existing empathy evaluation frameworks [28], [47]. The final empathy score, S_{EMP}, is the average of these ratings, normalized across all N queries:

$$S_{\text{EMP}} = \frac{1}{5N} \sum_{n=1}^{N} s^i_{\text{EMP}}, \tag{5}$$

where s^i_{EMP} is the empathy score for the ith sample.

4. Robustness: The final score, S_{ROB}, evaluates the conversational robustness of the doctor agent. A robust agent must consistently adhere to its designated role and interaction protocols. Deviations, such as misdirecting questions or breaking character, can compromise the simulation's integrity and lead the conversation astray. Therefore, the doctor agent must demonstrate a high degree of fidelity to its underlying instructions to ensure a reliable and predictable evaluation process. This metric aligns with the clinical competency of Professionalism and Reliability [48], as defined by medical education standards like the ACGME core competencies. Just as a human clinician must maintain professional boundaries and follow institutional protocols, an AI agent must operate within its defined operational constraints to be considered clinically viable.

We quantify robustness by identifying and counting failure cases during the simulation. The failure modes listed below represent the most frequent types of dialogue breakdown observed during our evaluation.

- **Role-Breaking**: The doctor agent self-simulates the entire interaction, playing the roles of other agents instead of interacting with them.
- **Abrupt Termination**: The conversation ends prematurely (defined as ≤ 3 turns) due to the doctor agent's failure to use the required `<patient>` or `<clinical>` tags, or due to an improperly placed `</end>` tag.
- **Invalid Answer**: After a seemingly normal conversation, the doctor agent fails to provide a valid answer to the final question (e.g., by providing an answer not listed in the options).

In our evaluation, these failure modes often manifest concurrently. For instance, a doctor agent might break character by self-simulating the roles of the patient or clinical staff, which leads to an immediate and improper termination of the dialogue before any actual multi-agent interaction can occur.

The robustness score, S_{ROB}, is then calculated as the proportion of successful conversations:

$$S_{\text{ROB}} = 1 - \frac{n_{\text{Fail}}}{N} \tag{6}$$

where n_{Fail} is the total count of the failure cases described above.

IV. Methods

A. Datasets

To evaluate the performance of each LLM as a clinical conversational agent using the AutoMedic framework, we utilized six distinct medical QA datasets.

MedBullets: This dataset [49] comprises 308 multiple-choice questions formatted in the style of the USMLE, each with five answer options.

MedQA: Another USMLE-style medical QA benchmark, MedQA [50] contains 1,273 multiple-choice questions, each presenting five answer options.

MedXpertQA: MedXpertQA [51] is a medical QA benchmark featuring both textual and multimodal questions. Our evaluation focused exclusively on its text-based subset, which consists of 2,450 questions, each with ten answer choices.

MedMCQA: MedMCQA [52] is a multiple-choice question answering dataset derived from real-world medical entrance exams. For our experiments, we used its validation split, which provides the correct answers. After filtering to include only single-choice questions, our subset contained 2,816 questions, each with four answer options.

HEAD-QA: This multi-choice healthcare dataset [53] includes medical questions in both Spanish and English. We specifically used the test split of its English subset, which provides 2,742 questions, each with four answer options.

MMLU-Pro: MMLU-Pro [54] is a multi-task understanding dataset designed for rigorous LLM benchmarking. From this dataset, we extracted questions pertaining only to the health and biology categories, totalling 1,535 questions with three to ten answer options each.

For the MedQA and MedXpertQA datasets, we sampled 300 instances that are applicable with our framework for efficiency.

B. Models

For the doctor agent, we evaluate the performance of 9 distinct LLMs.These models are grouped into three categories.

Open-Source General LLMs: We selected a range of general-purpose open-source models. We utilized the instruction-tuned versions of Llama 3 [4] and Qwen3 [7], which feature 70B and 32B parameters, respectively. As a representative reasoning model, we included the 70B distilled version of DeepSeek-R1 [5]. We also used gpt-oss [55], another open-source reasoning model from OpenAI, in its 120B parameter version.

Open Source Biomedical LLMs: To assess domain-specific models, we employed two LLMs designed for the biomedical field: Med42-v2 [11] and OpenBioLLM [12], both

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TABLE I
HUMAN EXPERT EVALUATION RESULTS FOR PATIENT PROFILE GENERATION.

Filtering Accuracy of the Profile Generator	
Fleiss' Kappa	0.7630 ($p < 0.05$)
Cohen's Kappa	0.8197 ($p < 0.05$)
Quality of Generated Patient Profiles	
Percent Agreement	0.8992 ($p < 0.05$)
ICC(2,k)	0.8434 ($p < 0.05$)
Average Score	3.464

Notes: For the filtering accuracy assessment, Fleiss' Kappa quantifies inter-rater agreement among human experts, while Cohen's Kappa measures agreement between the experts' majority vote and the profile generator. For the quality evaluation of generated patient profiles, the percent agreement and intraclass correlation coefficient (ICC) indicate human expert reliability, and the "Average Score" represents the mean expert rating across all samples (maximum possible score: 4 points).

with 70B parameters. We also included HuatuoGPT-o1 [10], a 72B parameter model specifically tailored for medical reasoning tasks.

Proprietary General LLMs: Finally, we evaluated two high-performing proprietary models: GPT-4o [2] and Claude Sonnet 4 [56].

For the other agents (profile generator, patient, and clinical staff), we employed GPT-4o to ensure a stable and consistent conversation simulation.

C. Human Evaluation

To validate the accuracy and effectiveness of our framework, we conducted three distinct human expert evaluations with the participation of four licensed medical professionals. These evaluations validate the clinical validity of the AutoMedic framework and the CARE metric by demonstrating their alignment with the judgments of medical experts. The institutional review board approved these human evaluation studies (IRB No. 4-2025-0982).

The first study assessed the accuracy of the profile generator's filtering mechanism. We randomly sampled 24 medical queries from the six datasets, comprising 12 queries classified as "appropriate" and 12 as "inappropriate" by the profile generator agent. The human experts then independently judged the suitability of each sample for patient profile generation. We measured inter-rater reliability among the experts using Fleiss' Kappa and the agreement between our agent's classification and the experts' majority vote using Cohen's Kappa.

The second study evaluated the quality of the generated patient profiles. Experts were asked to rate 21 randomly sampled profiles on a 4-point scale. The evaluation criteria included faithfulness to the original medical query, completeness of information, and the clinical plausibility of any imputed data. To measure inter-rater reliability, we calculated both the percent agreement and intraclass correlation coefficient (ICC), a metric robust to chance agreement. We then determined the average quality score across all profiles.

The final study aimed to correlate our automated CARE metric with human judgment. We generated a total of 75 simulation results, comprising 15 cases for each of the five models acting as the doctor agent on the MedQA dataset. In a blinded setup, the human experts rated each result on a 3-point scale across the four CARE dimensions: accuracy, conversational efficiency and strategy, empathy, and robustness. We measured inter-rater reliability using both percent agreement and ICC. Finally, we assessed the alignment between experts' average scores and the calculated CARE metric scores using the Pearson correlation coefficient. These metric calculations were performed on both the full MedQA dataset and the specific subset of 15 samples used for human evaluation.

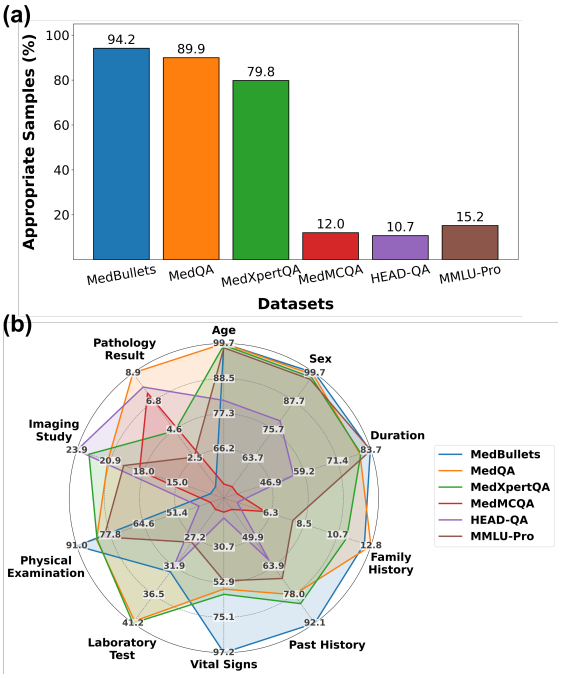


Fig. 3. Analysis of Medical QA Benchmark Datasets for Clinical Simulation. (a) A bar chart illustrates the proportion of "Appropriate Samples" within each dataset (calculated from the total samples). (b) A radar chart illustrates the percentage of appropriate samples that contain specific demographic, basic, and optional information for patient profile generation (calculated from the appropriate samples).

V. RESULTS

A. Human Evaluation of Patient Profile Generation.

To justify the patient profile generation process, we conducted two human expert evaluations. The first specifically assessed the profile generator's filtering accuracy. Human experts demonstrated substantial agreement on the suitability of queries for profile generation, indicated by a Fleiss' Kappa of 0.7630 (TABLE I). This high inter-rater reliability confirms that our filtering criteria are well-defined and consistently applied. Furthermore, the agreement between the profile generator's classifications and the human experts' majority vote was almost perfect, yielding a Cohen's Kappa of 0.8197 (excluding two tied cases). This evaluation robustly demonstrates that our profile generator accurately identifies and filters out inappropriate medical queries before patient profile generation.

Next, the evaluation of the generated patient profiles further supported our framework's design. Human expert ratings showed strong reliability, evidenced by a percent agreement of 0.8992 and an ICC of 0.8434 (TABLE I). This indicates consistent expert judgment. Additionally, the profiles received

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TABLE II
ALIGNMENT OF HUMAN EXPERT AND CARE METRIC EVALUATION FOR LLMS AS DOCTOR AGENTS.

		Accuracy	Conversational Efficiency & Strategy	Empathy	Robustness
Percent Agreement		0.9644 ($p < 0.05$)	0.7741 ($p < 0.05$)	0.8919 ($p < 0.05$)	0.7044 ($p < 0.05$)
ICC(2, k)		0.9839 ($p < 0.05$)	0.6083 ($p < 0.05$)	0.7207 ($p < 0.05$)	0.5281 ($p < 0.05$)
Average Human Score	Llama 3	1.6333	2.0167	2.3667	2.7333
	DeepSeek-R1	1.5500	1.8500	2.2167	2.2333
	Med42-v2	1.5333	1.8667	2.3167	1.8333
	HuatuoGPT-o1	2.2333	1.9000	2.7333	2.8167
	Claude Sonnet 4	2.2333	2.5000	2.9333	2.6500
CARE Metric (All Samples)	Llama 3	0.2865	0.6938	0.7260	1.0000
	DeepSeek-R1	0.0589	0.2073	0.6493	0.5867
	Med42-v2	0.0204	0.1350	0.6647	0.2700
	HuatuoGPT-o1	0.4095	0.4344	0.7493	0.9567
	Claude Sonnet 4	0.4699	0.7743	0.8520	0.9333
	Pearson Correlation	0.9061 ($p < 0.05$)	0.7953 ($p = 0.1077$)	0.9423 ($p < 0.05$)	0.9906 ($p < 0.05$)
CARE Metric (15 Samples)	Llama 3	0.1667	0.5442	0.6667	1.0000
	DeepSeek-R1	0.0222	0.1746	0.6133	0.5333
	Med42-v2	0.0067	0.0776	0.6267	0.2000
	HuatuoGPT-o1	0.5128	0.4717	0.7467	0.9333
	Claude Sonnet 4	0.4154	0.8671	0.8000	0.9333
	Pearson Correlation	0.9737 ($p < 0.05$)	0.8814 ($p < 0.05$)	0.9929 ($p < 0.05$)	0.9867 ($p < 0.05$)

Notes: This table presents inter-rater reliability metrics (Percent Agreement and Intraclass Correlation Coefficient, ICC), average human expert scores (3-point Likert scale), and corresponding CARE metric scores for five models across four evaluation dimensions when acting as a doctor agent on the MedQA dataset. Pearson correlation coefficients are provided to demonstrate the alignment between human expert ratings and CARE metric scores across four dimensions. "All Samples" indicates CARE metric calculations performed on the full MedQA dataset, while "15 Samples" refers to the specific subset subjected to human expert evaluation.

an average score of 3.464 out of a possible 4 points. This high average score suggests that the patient profiles generated by our framework are generally accurate and clinically reliable. These evaluations collectively validate the effectiveness of our patient profile generation process.

B. Dataset Suitability for Clinical Simulation

Before evaluating the performance of LLMs with our AutoMedic framework, we conducted a pre-analysis of the selected medical QA benchmarks to assess their suitability for simulating realistic clinical situations. Leveraging our validated patient profile generation process, we analyzed each dataset based on its capacity to provide sufficient and relevant information for virtual patient profiles. The detailed findings are presented in Fig. 3.

Our analysis first focused on the proportion of samples deemed appropriate for generating a virtual patient profile. As illustrated in Fig. 3(a), MedBullets and MedQA exhibited the highest proportion, with over 90% of their samples being suitable. MedXeprtQA, while having a slightly lower ratio of appropriate samples (79.8%), still contributed the largest absolute number of appropriate samples (1,955). In contrast, MedMCQA, HEAD-QA, and MMLU-Pro showed a significantly lower proportion of appropriate samples, ranging from 10% to 15%. This initial filter is crucial, as only appropriate queries can lead to meaningful clinical simulations.

Next, we assessed the informational richness of the appropriate samples, specifically evaluating whether the datasets provided sufficient details to construct comprehensive patient profiles. As shown in Fig. 3(b), demographics, such as age and sex, were generally well-covered across most datasets, though MedMCQA provided this information for only about half of its appropriate samples. For basic clinical information like symptom duration and past medical history, MedBullets, MedQA, and MedXpertQA consistently included these details for over 70% of their samples. Conversely, MedMCQA and HEAD-QA showed noticeable gaps, providing this information for a relatively smaller fraction of their queries. This disparity became even more pronounced when examining optional information, such as vital signs or physical examination findings. MedMCQA and HEAD-QA again presented the lowest ratios of inclusion. Interestingly, MMLU-Pro, despite its low proportion of appropriate samples, provided patient information with a frequency similar to the more suitable datasets once a sample was deemed appropriate.

In summary, MedBullets, MedQA, and MedXpertQA emerge as the most suitable datasets for the AutoMedic framework. They not only offer a high proportion (or a large absolute number) of appropriate samples for patient profile generation but also provide abundant and comprehensive information to create detailed virtual patient profiles, essential for accurate clinical situation simulation. MedBullets and MedQA stand out for their high proportional suitability, while MedXpertQA excels in the total volume of usable samples.

Conversely, MedMCQA and HEAD-QA, despite potentially having large total numbers of medical queries, are less optimal for our framework due to their low proportion of appropriate samples and their subsequent lack of detailed information necessary for robust patient profile generation. MMLU-Pro, while also having a low proportion of appropriate samples, is somewhat mitigated by the relatively good information completeness when a sample is appropriate, suggesting its suitable samples are of higher quality.

C. Correlation of CARE Metric with Human Expert Evaluations

Before deploying the CARE metric for LLM evaluation, we first established its alignment with the human expert judgment.

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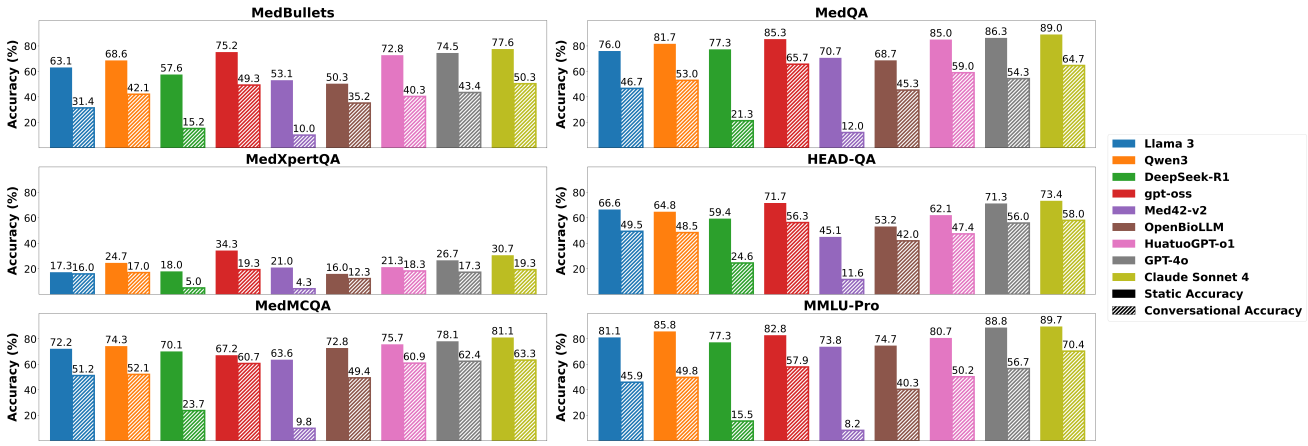


Fig. 4. Static and conversational QA accuracy of LLMs across medical QA benchmarks. This figure presents the static QA accuracy (solid bars) and conversational QA accuracy (striped bars) for each LLM across the six medical QA benchmark datasets.

TABLE II presents the results of human expert evaluation for five models acting as doctor agents on a random 15 samples from the MedQA dataset, alongside their corresponding CARE metric scores.

For accuracy, human experts demonstrated very high reliability, with an ICC value of 0.9839. This high agreement likely stems from the relatively objective nature of judging the correctness of a medical answer. DeepSeek-R1 and Med42-v2 received relatively low average scores of approximately 1.5, whereas HuatuoGPT-o1 and Claude Sonnet 4 achieved notably higher average scores of 2.2333. This trend is directly reflected in the CARE metric results, where DeepSeek-R1 and Med42-v2 recorded lower S_{ACC} values, while HuatuoGPT-o1 and Claude Sonnet 4 exhibited higher S_{ACC} scores. Furthermore, the Pearson correlation coefficients between the average human scores and the CARE metric were high for both the full dataset (0.9061) and the 15-sample subset (0.9737), indicating a strong alignment between automated and expert evaluation.

Regarding conversation efficiency and strategy, the human expert agreement, while lower than for accuracy, remained moderate, yielding an ICC value of 0.6083. Human evaluators assigned Claude Sonnet 4 the highest average score of 2.5000 and Llama 3 a score of 2.0167, while all other models received average scores below 2.0. This performance hierarchy is consistently mirrored by the S_{CES} component of the CARE metric. While the Pearson correlation between human ratings and the CARE metric calculated on the full dataset was 0.7953 ($p = 0.1077$), the correlation increased to 0.8814 with a statistically significant p-value ($p < 0.05$) when the CARE metric was calculated specifically for the 15-sample subset. This suggests that the automated metric effectively captures the nuances of conversational flow and strategy perceived by human experts.

Regarding empathy, human evaluators exhibited high agreement, with a percent agreement of 0.8919 and an ICC value of 0.7207. Claude Sonnet 4 achieved a near-perfect average human score of 2.9333, significantly outperforming most other models with the exception of HuatuoGPT-o1, which received a score of 2.7333. This hierarchy is consistently mirrored by the S_{EMP} component of the CARE metric, where Claude Sonnet 4 recorded the highest empathy scores, specifically 0.8520 on the full dataset and 0.8000 on the 15-sample subset. Notably, the Pearson correlation for empathy on the 15-sample subset was the highest among all four evaluation dimensions, further reinforcing the strong alignment between the automated metric and human judgment.

Finally, regarding robustness, human experts demonstrated moderate agreement with an ICC value of 0.5281 with ($p < 0.05$). In terms of average human scores, HuatuoGPT-o1 received the highest rating of 2.8167, while Med42-v2 recorded the lowest score of 1.8333. Although Llama 3 achieved the maximum CARE metric score for robustness (1.0000), the overall performance trends of the CARE metric remained closely aligned with those observed in the human evaluations. Furthermore, the Pearson correlation coefficients for the full dataset and the 15-sample subset were 0.9906 and 0.9867, respectively, indicating a very strong alignment between human expert judgement and the CARE metric.

In summary, the consistent inter-rater reliability observed across all four CARE dimensions underscores the robustness of the human expert evaluations. The trends in average human scores demonstrate a strong alignment with the corresponding CARE metric values across diverse LLMs. This robust correlation validates the CARE metric as an appropriate and reliable tool for assessing the multifaceted capabilities of LLMs acting as clinical conversational agents.

D. Accuracy in Static and Conversational Settings

While recent studies have observed that the performance of LLMs often degrades during multi-turn interactions [57], this phenomenon remains insufficiently quantified within the clinical domain. To systematically measure this performance gap, we first comprehensively compare the performance of LLMs by examining their accuracy in both static QA and conversational settings. Fig. 4 illustrates these accuracies across the various medical QA datasets.

A clear trend emerges whereby models exhibiting high QA accuracy generally maintain relatively high conversational accuracy. However, a consistent observation across all experiments is that every model demonstrates lower performance in the conversational setting compared to its static

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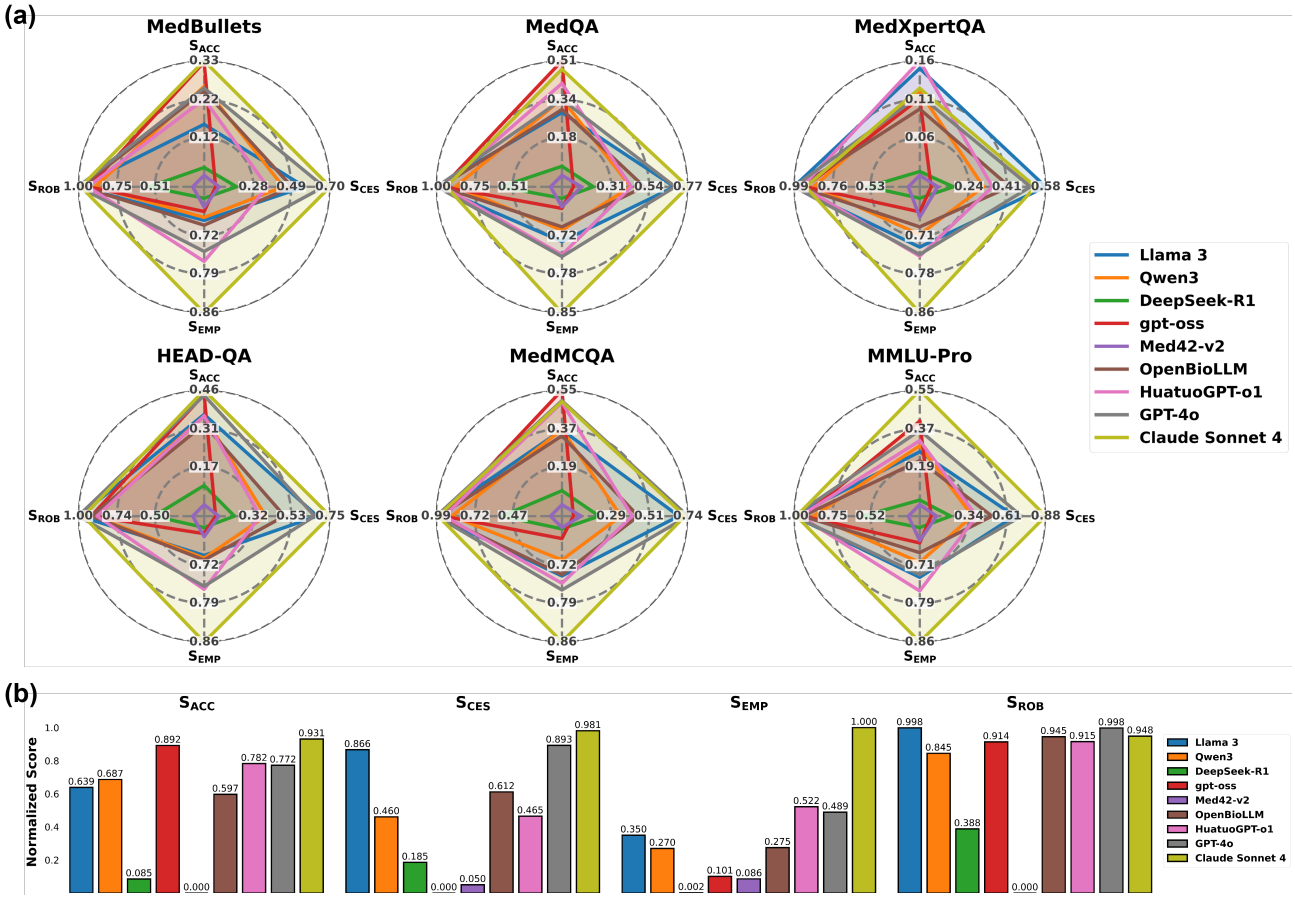


Fig. 5. (a) LLM performance across medical QA benchmarks via CARE metric. These radar charts display the performance of various LLMs on each medical QA dataset, illustrating scores across the four CARE metric dimensions: S_{ACC} (accuracy), S_{CES} (conversational efficiency & strategy), S_{EMP} (empathy), and S_{ROB} (robustness). (b) Average normalized CARE metric across medical QA benchmarks. This bar plot illustrates the average normalized scores for each component of the CARE metric (S_{ACC}, S_{CES}, S_{EMP}, S_{ROB}) across all evaluated medical QA datasets. Scores are min-max normalized per dataset, with 1 representing the highest score and 0 the lowest for each metric.

QA counterpart. This performance drop is particularly pronounced for DeepSeek-R1 and Med42-v2, as evident in their significant accuracy decline in conversational scenarios across most datasets. This universal struggle suggests that models, regardless of their underlying knowledge, face substantial challenges in effectively gathering all necessary information within a dynamic conversational exchange.

This finding provides a critical insight that an LLM's raw knowledge in medicine (reflected in QA accuracy) does not directly translate to effective clinical conversational ability. The conversational context introduces complexities such as managing dialogue flow, interpreting implicit cues, maintaining coherence, and adapting to user responses, all of which are distinct from simply answering a well-posed, single-turn question. Therefore, evaluating LLMs specifically as clinical conversational agents, rather than solely on static QA, is essential. This approach reveals whether models can effectively communicate with users, even if they possess extensive domain knowledge, highlighting the importance of metrics that capture interactive performance.

E. Clinical Conversation Performance Analysis using the CARE Metric

Next, we evaluated the capabilities of various LLMs as clinical conversational agents using the comprehensive CARE metric. Fig. 5(a) visually summarizes these evaluation results across the medical QA datasets.

Among the open-source general models, Llama 3 and Qwen3 exhibited average performance across the overall CARE metric. Llama3 notably achieved the highest S_{ROB} across the board, though the inter-model differences in robustness were generally minor. Qwen3 demonstrated strong S_{ACC}, yet it did not surpass larger models like gpt-oss or the proprietary models. DeepSeek-R1 consistently performed poorly across most experiments. DeepSeek-R1 struggled to function appropriately within a virtual clinical setting (e.g., ended the conversation inappropriately). We hypothesize this degradation in performance stems from its distilled nature, where supervised fine-tuning on synthetic data, rather than reinforcement learning, may have impaired both its reasoning capabilities and its ability to engage in proper communication with other agents, a phenomenon observed in prior work [58]. gpt-oss achieved high S_{ACC} due to strong QA and conversational accuracy, but received low scores for both S_{CES} and S_{EMP}. This deficiency in conversational strategy and empathy

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is due to the model's reliance on uncharacteristic formats, such as numbered lists, a style that emphasizes objective communication over human interaction. This detached, objective approach inherently ensures the absence of sycophancy.

Turning to open-source biomedical LLMs, Med42-v2 generally performed poorly across most CARE metrics. Despite being fine-tuned on medical data, it showed no significant improvement in S_{ACC} and underperformed in other dimensions.

Moreover, it exhibited abnormal conversational behavior, such as answering its own questions without waiting for input from the patient or clinical staff. OpenBioLLM achieved better S_{CES} and S_{ROB} compared to Med42-v2, but it too exhibited degraded performance in accuracy and empathy. In contrast, HuatuoGPT-o1 demonstrated superior performance with the exception of S_{CES}. This lower efficiency stems from its tendency to repeatedly request the same test results from the clinical staff agent, even after being informed that those results are unavailable. Nevertheless, its S_{ACC} scores were comparable to those of gpt-oss and proprietary models, which we attribute to its combination of medical domain-specific tuning and robust reasoning abilities.

The two proprietary general LLMs presented different strengths. Claude Sonnet 4 achieved a high S_{ACC} due to its excellent QA and conversational accuracy, with a minimal gap between the two. Furthermore, it recorded the highest S_{EMP} across all models, consistently exceeding 0.85 on all datasets. This superior empathetic performance is likely due to its tendency to check patient status and express empathy, unlike other LLMs that primarily focus on information gathering. GPT-4o, while showing relatively lower S_{ACC} and S_{EMP} compared to Claude Sonnet 4, slightly outperformed it in S_{ROB}. This indicates that GPT-4o possesses superior robustness, effectively maintaining its assigned persona and strictly adhering to conversational protocols, such as the consistent use of appropriate tags.

To enable a comprehensive comparison of model performance across diverse datasets, we applied min-max normalization for each metric within each dataset (where 1 represents the highest score and 0 the lowest). Subsequently, these normalized scores were averaged across all datasets for each metric. Fig. 5(b) presents these average normalized overall CARE metric scores.

For S_{ACC}, Claude Sonnet 4 demonstrates the strongest performance, closely followed by gpt-oss. In terms of S_{CES}, Claude Sonnet 4 achieves the highest average score, with GPT-4o securing the second position. Regarding S_{EMP}, Claude Sonnet 4 stands out with an overwhelmingly superior performance, indicating its exceptional ability in this dimension. For S_{ROB}, Llama 3 and GPT-4o lead, although the performance differences among models in this category are relatively small. Notably, DeepSeek-R1 and Med42-v2 consistently exhibit poor performance across most CARE metrics. These specific trends will be discussed in greater detail in the subsequent discussion section.

VI. Discussion

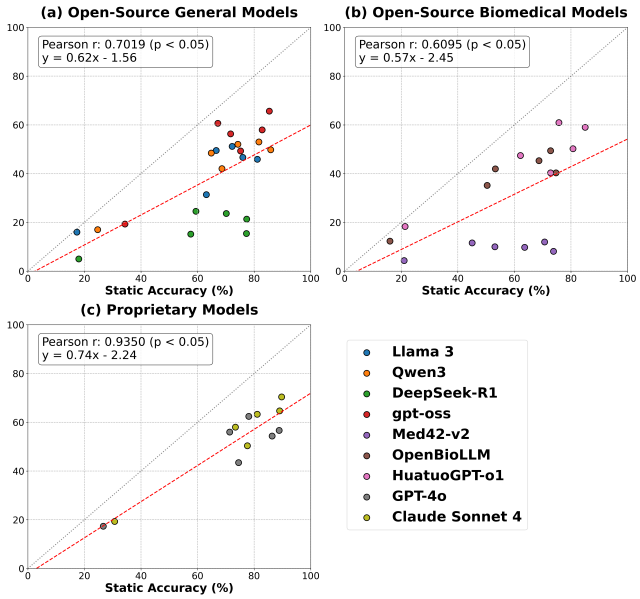


Fig. 6. The relationship between static and conversational QA accuracy. The scatter plots display the correlation for: (a) Open-source general models, (b) Open-source biomedical models, and (c) Proprietary models. For each category, the Pearson correlation coefficient (r), p-value, and linear regression equation are provided in the top-left corner. The grey dashed line represents the identity line ($y = x$), while the red dashed line indicates the fitted regression line.

A. Discussion

Leveraging the patient profile generation capabilities of AutoMedic, we first established guidelines for assessing the suitability of medical QA datasets for clinical simulation. Our analysis revealed significant disparities among datasets regarding the volume of clinically appropriate samples and the depth of patient information provided. We found that datasets such as MedBullets, MedQA, and MedXpertQA offer a substantial number of appropriate samples enriched with sufficient patient details. In contrast, benchmarks like MedMCQA and HEAD-QA exhibit a low proportion of suitable samples and frequently lack basic patient information. These results imply that not all medical QA datasets contain questions relevant to realistic clinical situations. Therefore, relying solely on them to assess the clinical capabilities of LLMs may be inadequate. Consequently, our framework serves as a vital guideline for determining the applicability of medical QA datasets to clinical scenario simulations. Furthermore, a key advantage of our framework is its generalizability. Although we utilized six specific datasets in this study, AutoMedic is designed to automatically adapt to and process any new off-the-shelf medical QA dataset without manual modification.

We compared the performance in both static QA and conversational settings, revealing that the dynamic conversational environment presents a significantly greater challenge than static QA. To quantify the relationship between these two performance modes, we performed a linear regression analysis and calculated the Pearson correlation coefficient (r). When examining open-source general models (Fig. 6(a)), a strong correlation is observed, although DeepSeek-R1 notably deviates from the general trend. In contrast, open-source biomedical models (Fig. 6(b)) exhibit the weakest correla-

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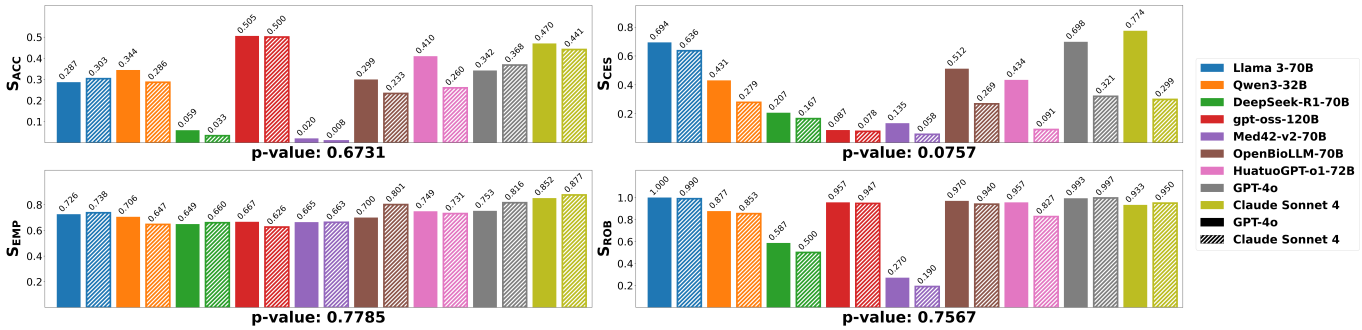


Fig. 7. Impact of base models for patient/clinical staff agents on CARE metrics with the MedQA dataset. The bar charts illustrate the component score S_{ACC}, S_{CES}, S_{EMP}, and S_{ROB} for each evaluated model. Solid bars indicate GPT-4o as the base model for the patient/clinical staff agents, while striped bars indicate Claude Sonnet 4. The p-values are obtained from a one-way ANOVA test to assess the statistical consistency of the results across different base model configurations.

tion ($r = 0.6095$). These results suggest that while static medical knowledge serves as a foundation for conversational competence, suboptimal fine-tuning strategies may hinder the effective translation of this knowledge into clinical dialogue. Conversely, proprietary models (Fig. 6(c)) demonstrate the highest correlation ($r = 0.9350$), indicating a consistent balance between their static clinical knowledge and conversational capabilities. In conclusion, for clinical conversational settings, it is advantageous to prioritize proprietary models or carefully select open-source models that demonstrate both high static QA accuracy and robust instruction-following tuning.

Our evaluation using the CARE metric revealed that domain-specific medical tuning does not necessarily confer superiority in conversational settings. With the notable exception of HuatuoGPT-o1, most medical-tuned models underperformed compared to their general-domain counterparts. We conjecture that this performance deficit arises because these models were primarily fine-tuned for static medical QA tasks rather than dynamic, interactive communication. During such specialized fine-tuning, models may inadvertently lose their broader communicative capabilities. Consequently, they fail to leverage their specialized knowledge effectively in realistic clinical scenarios because they cannot manage the conversational flow. This result indicates that fine-tuning with domain-specific data must be approached carefully to ensure that foundational communication skills are preserved. It also highlights a significant generalization gap where static medical expertise does not automatically translate into effective interactive reasoning.

A comparison between open-source and proprietary models revealed significant differences in their performance profiles. Open-source models often demonstrated "spiky" capabilities, excelling in specific dimensions while underperforming in others. For instance, gpt-oss achieved the high S_{ACC} and Llama 3 led in S_{ROB}, but both showed considerable deficits in other metrics. In contrast, the proprietary models, GPT-4o and Claude Sonnet 4, exhibited a more balanced and consistently high performance across all CARE dimensions. Although they did not always secure the highest score in every individual metric, their strong, well-rounded profiles suggest they are currently the most reliable choice for a comprehensive clinical conversational agent. However, in scenarios where proprietary models are unfeasible due to data privacy constraints, data export limitations, or other restrictions, our findings indicate that HuatuoGPT-o1 presents itself as the most viable open-source alternative.

Beyond technical accuracy, the integration of LLMs into clinical practice requires rigorous adherence to ethical and safety standards. As highlighted in [59], clinical chatbots must be developed within a human-centered AI framework that prioritizes empathy, transparency, and equity. This includes addressing significant risks such as algorithmic bias, where models trained on non-representative datasets may provide suboptimal care to marginalized populations. Our proposed framework addresses these ethical imperatives by incorporating metrics such as empathy and conversational robustness. These rubrics evaluate the model's ability to provide supportive care and maintain professional interaction protocols. By quantifying these human-centric dimensions, the CARE metric facilitates a safer and more ethical deployment of AI in medicine.

B. Limitations and Future Works

While our framework provides a comprehensive evaluation of LLMs as clinical conversational agents, several limitations must be addressed. AutoMedic is currently limited to text-based interactions and objective multiple-choice outcomes. Future work will extend the framework to multimodal data by using the clinical staff agent as a gateway for medical imaging or reports. We also aim to incorporate free-text diagnostic generation evaluated by medical consensus models to capture the nuances of open-ended reasoning while balancing evaluation complexity. Furthermore, we intend to explore specialized training methodologies and collaborative strategies such as agent-based debate to support collective intelligence beyond our current focus on internal clinical logic.

While our human validation shows high correlation with expert judgment, the current scope is limited to specific models and the MedQA dataset. Future research should validate the framework across broader specialties, patient demographics, and personas to ensure clinical utility. We also acknowledge that the empathy metric serves as a proxy for perceived rapport within a simulation rather than a direct evaluation of a model's emotional state. Although this proxy aligns with expert judgment, further validation across diverse datasets and

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institutional settings is necessary to confirm broad generalizability.

As illustrated in Fig. 7, the behavior of the supporting agents can influence S_{CES} scores, which may introduce model-dependent bias and limit the framework's model-agnosticism. To minimize this, future work should utilize an ensemble of diverse models for the patient and clinical staff roles. We also recognize the S_{CES} Gaussian assumptions represent a normative design choice based on established primary care guidelines. These parameters are intentionally adjustable to allow calibration against diverse clinical contexts such as psychiatric intake or emergency triage. Moreover, future versions of S_{ROB} should expand from technical protocol compliance to explicitly assess clinical safety and hallucination risk. Adopting more flexible parsing could also improve our ability to disentangle core reasoning from technical compliance.

Finally, we did not define a single, aggregated score for overall performance because the CARE dimensions vary in scale and application-specific importance. Future research could develop weighted composite scores calibrated against human expert preferences from head-to-head model comparisons. We therefore present this methodology as an iterative foundation for automated assessment that will benefit from continued empirical testing.

VII. Conclusion

In this paper, we introduced AutoMedic, a novel, automated multi-agent framework that enables the evaluation of LLMs as clinical conversational agents through realistic, simulated dialogues. To quantitatively assess performance within this framework, we also proposed the CARE metric, a multi-faceted tool that measures accuracy, conversational efficiency and strategy, empathy, and robustness.

Our investigation yielded several key findings. Human expert evaluations confirmed the reliability of AutoMedic's patient profile generation and the strong correlation of our CARE metric with human clinical judgment. We found that only a subset of medical QA datasets, such as MedBullets, MedQA, and MedXpertQA, are suitable for clinical simulation. Crucially, we observed that all LLMs suffer a performance drop when moving from static QA to conversational settings. This finding confirms that existing static medical QA benchmarks, which primarily assess knowledge, do not sufficiently reflect the actual performance of LLMs in dynamic medical field applications. The CARE metric further revealed distinct LLM performance profiles. Proprietary models like Claude Sonnet 4 and GPT-4o showed balanced, high-level performance, while most medical-tuned models underperformed, suggesting fine-tuning may erode communication skills. HuatuoGPT-o1, however, emerged as a strong open-source alternative.

This work provides a robust, automated methodology for the comprehensive evaluation of conversational medical AI, offering deeper insights than static metrics alone. AutoMedic and the CARE metric can guide researchers and developers in building and selecting LLMs that are not only knowledgeable but also effective, empathetic, and reliable in practice. Furthermore, by providing a quantitative and fully automated evaluation framework for clinical conversational LLMs, our work facilitates the safe and rigorous integration of these models into real-world clinical practice. We believe that the AutoMedic framework and CARE metric offer a valuable guide for identifying appropriate models to serve in assistive or decision-support roles alongside human clinicians.

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